

V3 empirical study · Instruction

Five models · four underlyings · four volatility regimes · 1.5-year monthly calibration · return-based calibration

What this report is

- Computational source of truth: the companion Jupyter notebook. This PDF is written from the same payload; every table number is identical.
- Question: which of five models tracks realized stock price S_t most closely, and does the ranking change across companies and the four volatility regimes?
- This report does not price options. Option RMSE lives in the companion decision PDF.
- In every table the BEST row is the lowest percentage RMSE of the p50 path versus realized S_t . That row is bold and shaded green; Ranking ranks 1–5.
- Companion stock-path report. This file keeps MD-GBM, GBM, GARCH, Merton, and GARCH-Merton, whose P dynamics come from lookback returns. MD-GBM means Markov Directional Geometric Brownian Motion. Rankings are within this group; they are not compared here with Heston / Heston-Merton.
- The other stock companion is V4_1p5y_monthly_stock_price_option_implied.ipynb.

Experimental design (held fixed for every model)

- Models: MD-GBM (Markov Directional Geometric Brownian Motion), GBM, GARCH(1,1), Merton jump-diffusion, GARCH-Merton.
- Companies: SPY, AAPL, MSFT, AMZN. Each name is calibrated and simulated separately.
- Regimes: Crisis 2008-08-01→2009-07-31; Normal 2014-01-01→2014-12-31; Late-cycle 2018-10-01→2019-09-30; COVID 2019-09-01→2020-08-31.
- Calibration window: 1.5-year ending at each monthly update. Rolling: monthly (period start plus month-ends). Parameters estimated at t are used only after t (no look-ahead).
- P-measure: keep the lookback drift μ . Do not replace $\mu \rightarrow r$ (that replacement is only for LSM option pricing).

V3 empirical study · Instruction (continued)

Five models · four underlyings · four volatility regimes · 1.5-year monthly calibration · return-based calibration

Path construction

- One rolling path cloud per ticker $\times$ regime $\times$ model, $n_paths = 10000$, seed 42.
- Clock: daily trading days from the first bar of the regime through the regime end. $\Delta t = 1/252$.
- Start at the observed open S_0 . Refresh parameters at month-ends with data $\leq$ that date.
- At each day, summarize the cloud by p10, p25, p50, p75, p90. The reported path is p50, not the Monte Carlo mean.

Metrics

- Primary: $RMSE\% = 100 \times \sqrt{\text{mean}(((p50 - S_t)/S_t)^2)}$. Ranking uses this only.
- $MAE\% = 100 \times \text{mean}(|p50 - S_t|/S_t)$. $Bias\% = 100 \times \text{mean}((p50 - S_t)/S_t)$.
- Tables report RMSE%, MAE%, Bias%, and Ranking only.

Page 3 · Table block · SPY · five models × four regimes

P-measure daily cloud, reported path = p50. Bold green = lowest RMSE% vs realized S_t . 1.5-year lookback, rolling=monthly, n_paths=10000, seed=42.

Table SPY-1 SPY · 2008-2009 Crisis
2008-08-01 → 2009-07-31 · n = 252 days

Model	RMSE%	MAE%	Bias%	Ranking
MD-GBM	27.08	21.76	+21.12	2
GBM	25.80	20.35	+19.23	1
GARCH	33.58	28.91	+28.54	5
Merton	27.61	22.46	+21.84	3
GARCH-Merton	33.42	28.77	+28.34	4

Table SPY-2 SPY · 2013-2014 Normal
2014-01-01 → 2014-12-31 · n = 253 days

Model	RMSE%	MAE%	Bias%	Ranking
MD-GBM	4.67	3.91	+3.90	2
GBM	4.49	3.72	+3.72	1
GARCH	8.02	6.91	+6.91	5
Merton	7.02	5.94	+5.94	4
GARCH-Merton	5.37	4.56	+4.55	3

Table SPY-3 SPY · 2018-2019 Late-cycle
2018-10-01 → 2019-09-30 · n = 251 days

Model	RMSE%	MAE%	Bias%	Ranking
MD-GBM	9.21	8.05	+8.05	2
GBM	8.90	7.69	+7.69	1
GARCH	24.77	22.96	+22.96	5
Merton	13.33	12.63	+12.63	4
GARCH-Merton	12.22	11.55	+11.55	3

Table SPY-4 SPY · 2019-2020 COVID
2019-09-01 → 2020-08-31 · n = 252 days

Model	RMSE%	MAE%	Bias%	Ranking
MD-GBM	8.40	6.36	-1.25	1
GBM	8.57	6.67	-1.91	2
GARCH	20.26	14.62	+14.02	5
Merton	9.02	6.30	+0.60	3
GARCH-Merton	12.93	10.37	+5.68	4

RMSE% is $100 \times \sqrt{\text{mean}(((p50 - S_t)/S_t)^2)}$. MAE% is $100 \times \text{mean}(|p50 - S_t|/S_t)$. Bias% is $100 \times \text{mean}((p50 - S_t)/S_t)$. Ranking uses RMSE% only.

Page 4 · Table block · Company summary · five models × four regimes

Each number is the arithmetic mean of AAPL, MSFT, AMZN. Same columns as the SPY page. Bold green = lowest mean RMSE% vs realized S_t. 1.5-year lookahead, rolling=monthly, n_paths=

Table Co-1 Company summary · 2008-2009 Crisis

2008-08-01 → 2009-07-31 · mean of AAPL, MSFT, AMZN · $\bar{n}$ = 252 days

Model	RMSE%	MAE%	Bias%	Ranking
MD-GBM	39.57	30.27	+25.99	3
GBM	36.04	28.00	+19.79	2
GARCH	51.63	44.05	+41.08	4
Merton	35.73	28.75	+16.53	1
GARCH-Merton	56.38	48.80	+46.41	5

Table Co-2 Company summary · 2013-2014 Normal

2014-01-01 → 2014-12-31 · mean of AAPL, MSFT, AMZN · $\bar{n}$ = 253 days

Model	RMSE%	MAE%	Bias%	Ranking
MD-GBM	19.41	17.31	+7.00	4
GBM	19.27	17.15	+5.06	3
GARCH	17.99	16.31	+7.93	1
Merton	18.87	16.73	+10.39	2
GARCH-Merton	20.87	18.68	+6.41	5

Table Co-3 Company summary · 2018-2019 Late-cycle

2018-10-01 → 2019-09-30 · mean of AAPL, MSFT, AMZN · $\bar{n}$ = 251 days

Model	RMSE%	MAE%	Bias%	Ranking
MD-GBM	28.46	26.36	+26.35	3
GBM	26.51	24.33	+24.32	2
GARCH	50.42	46.52	+46.51	5
Merton	25.40	23.03	+23.02	1
GARCH-Merton	40.72	37.55	+37.54	4

Table Co-4 Company summary · 2019-2020 COVID

2019-09-01 → 2020-08-31 · mean of AAPL, MSFT, AMZN · $\bar{n}$ = 252 days

Model	RMSE%	MAE%	Bias%	Ranking
MD-GBM	17.66	14.50	-13.26	3
GBM	19.09	15.67	-14.78	5
GARCH	9.81	7.89	-0.47	1
Merton	18.69	15.22	-14.20	4
GARCH-Merton	11.20	9.40	-3.85	2

RMSE%, MAE%, and Bias% are each averaged across AAPL, MSFT, AMZN. This is not a pooled re-simulation of a combined price series. Ranking uses the mean RMSE% only.

Page 5 · Table block · AAPL · five models × four regimes

P-measure daily cloud, reported path = p50. Bold green = lowest RMSE% vs realized S_t . 1.5-year lookback, rolling=monthly, n_paths=10000, seed=42.

Table AAPL-1 AAPL · 2008-2009 Crisis
2008-08-01 → 2009-07-31 · n = 252 days

Model	RMSE%	MAE%	Bias%	Ranking
MD-GBM	57.33	46.16	+44.38	3
GBM	51.72	40.81	+37.11	1
GARCH	88.07	79.08	+78.06	4
Merton	56.69	44.80	+42.41	2
GARCH-Merton	99.38	89.70	+88.82	5

Table AAPL-2 AAPL · 2013-2014 Normal
2014-01-01 → 2014-12-31 · n = 253 days

Model	RMSE%	MAE%	Bias%	Ranking
MD-GBM	16.29	13.94	-12.31	4
GBM	18.03	15.26	-14.06	5
GARCH	13.30	11.75	-9.22	2
Merton	11.21	10.07	-6.75	1
GARCH-Merton	14.97	12.99	-10.92	3

Table AAPL-3 AAPL · 2018-2019 Late-cycle
2018-10-01 → 2019-09-30 · n = 251 days

Model	RMSE%	MAE%	Bias%	Ranking
MD-GBM	32.15	28.78	+28.76	3
GBM	30.71	27.14	+27.12	2
GARCH	46.80	43.80	+43.78	5
Merton	29.96	26.36	+26.34	1
GARCH-Merton	40.39	37.68	+37.66	4

Table AAPL-4 AAPL · 2019-2020 COVID
2019-09-01 → 2020-08-31 · n = 252 days

Model	RMSE%	MAE%	Bias%	Ranking
MD-GBM	24.58	22.00	-21.96	4
GBM	26.10	23.43	-23.41	5
GARCH	10.87	9.11	-3.36	1
Merton	24.55	21.92	-21.87	3
GARCH-Merton	11.40	9.42	-7.05	2

RMSE% is $100 \times \sqrt{\text{mean}(((p50 - S_t)/S_t)^2)}$. MAE% is $100 \times \text{mean}(|p50 - S_t|/S_t)$. Bias% is $100 \times \text{mean}((p50 - S_t)/S_t)$. Ranking uses RMSE% only.

Page 6 · Table block · MSFT · five models × four regimes

P-measure daily cloud, reported path = p50. Bold green = lowest RMSE% vs realized S_t . 1.5-year lookback, rolling=monthly, n_paths=10000, seed=42.

Table MSFT-1 MSFT · 2008-2009 Crisis
2008-08-01 → 2009-07-31 · n = 252 days

Model	RMSE%	MAE%	Bias%	Ranking
MD-GBM	19.39	16.24	+9.55	4
GBM	18.26	15.35	+6.51	2
GARCH	18.01	15.09	+8.32	1
Merton	18.66	15.83	+4.33	3
GARCH-Merton	20.00	16.39	+11.42	5

Table MSFT-2 MSFT · 2013-2014 Normal
2014-01-01 → 2014-12-31 · n = 253 days

Model	RMSE%	MAE%	Bias%	Ranking
MD-GBM	3.46	2.88	-1.80	3
GBM	4.49	3.85	-3.08	4
GARCH	3.23	2.65	-1.51	2
Merton	2.73	2.11	-0.04	1
GARCH-Merton	8.09	6.81	-6.08	5

Table MSFT-3 MSFT · 2018-2019 Late-cycle
2018-10-01 → 2019-09-30 · n = 251 days

Model	RMSE%	MAE%	Bias%	Ranking
MD-GBM	13.52	12.21	+12.21	3
GBM	12.44	10.84	+10.84	2
GARCH	25.64	24.40	+24.40	5
Merton	12.29	10.16	+10.16	1
GARCH-Merton	17.28	16.42	+16.42	4

Table MSFT-4 MSFT · 2019-2020 COVID
2019-09-01 → 2020-08-31 · n = 252 days

Model	RMSE%	MAE%	Bias%	Ranking
MD-GBM	9.26	7.59	-6.14	3
GBM	10.72	8.82	-7.67	4
GARCH	6.27	4.05	+1.88	1
Merton	10.88	9.06	-8.18	5
GARCH-Merton	8.38	7.02	-5.84	2

RMSE% is $100 \times \sqrt{\text{mean}(((p50 - S_t)/S_t)^2)}$. MAE% is $100 \times \text{mean}(|p50 - S_t|/S_t)$. Bias% is $100 \times \text{mean}((p50 - S_t)/S_t)$. Ranking uses RMSE% only.

Page 7 · Table block · AMZN · five models × four regimes

P-measure daily cloud, reported path = p50. Bold green = lowest RMSE% vs realized S_t . 1.5-year lookback, rolling=monthly, n_paths=10000, seed=42.

Table AMZN-1 AMZN · 2008-2009 Crisis
2008-08-01 → 2009-07-31 · n = 252 days

Model	RMSE%	MAE%	Bias%	Ranking
MD-GBM	41.99	28.41	+24.03	3
GBM	38.14	27.84	+15.74	2
GARCH	48.79	38.00	+36.87	4
Merton	31.83	25.62	+2.85	1
GARCH-Merton	49.75	40.30	+38.98	5

Table AMZN-2 AMZN · 2013-2014 Normal
2014-01-01 → 2014-12-31 · n = 253 days

Model	RMSE%	MAE%	Bias%	Ranking
MD-GBM	38.47	35.12	+35.11	3
GBM	35.30	32.34	+32.32	1
GARCH	37.43	34.53	+34.53	2
Merton	42.68	38.00	+37.97	5
GARCH-Merton	39.54	36.24	+36.22	4

Table AMZN-3 AMZN · 2018-2019 Late-cycle
2018-10-01 → 2019-09-30 · n = 251 days

Model	RMSE%	MAE%	Bias%	Ranking
MD-GBM	39.72	38.08	+38.08	3
GBM	36.37	34.99	+34.99	2
GARCH	78.80	71.35	+71.35	5
Merton	33.94	32.57	+32.57	1
GARCH-Merton	64.50	58.55	+58.55	4

Table AMZN-4 AMZN · 2019-2020 COVID
2019-09-01 → 2020-08-31 · n = 252 days

Model	RMSE%	MAE%	Bias%	Ranking
MD-GBM	19.14	13.91	-11.69	3
GBM	20.44	14.77	-13.27	4
GARCH	12.30	10.52	+0.06	1
Merton	20.65	14.66	-12.56	5
GARCH-Merton	13.83	11.75	+1.36	2

RMSE% is $100 \times \sqrt{\text{mean}(((p50 - S_t)/S_t)^2)}$. MAE% is $100 \times \text{mean}(|p50 - S_t|/S_t)$. Bias% is $100 \times \text{mean}((p50 - S_t)/S_t)$. Ranking uses RMSE% only.

Page 7 · Summary tables · RMSE% across companies and regimes

Table S1 · Overall ranking (mean RMSE% over 16 cells)

Rank	Model	Mean RMSE%	Median RMSE%	# of 16 cells best
1	GBM	21.90	19.35	5
2	Merton	22.06	19.65	6
3	MD-GBM	22.79	19.26	1
4	GARCH-Merton	28.21	16.12	0
5	GARCH	29.76	22.52	4

Table S2 · Wins by company (lowest RMSE% in that company×regime)

Model	SPY wins	AAPL wins	MSFT wins	AMZN wins	Total
MD-GBM	1	0	0	0	1
GBM	3	1	0	1	5
GARCH	0	1	2	1	4
Merton	0	2	2	2	6
GARCH-Merton	0	0	0	0	0

Table S3 · Mean RMSE% by regime (equal-weight mean of SPY / AAPL / MSFT / AMZN)

Model	2008-2009 Crisis	2013-2014 Normal	2018-2019 Late-cycle	2019-2020 COVID	Mean of 4
MD-GBM	36.45	15.72	23.65	15.34	22.79
GBM	33.48	15.58	22.11	16.46	21.90
GARCH	47.11	15.49	44.00	12.43	29.76
Merton	33.70	15.91	22.38	16.28	22.06
GARCH-Merton	50.64	16.99	33.59	11.63	28.21

Green row in S1 / S3 = lowest mean RMSE%. Green row in S2 = most company×regime wins. A model can win the most cells without having the lowest average error (and vice versa).

Table S4.1 · SPY

Model	Mean RMSE%	Median	# regimes best
MD-GBM	12.34	8.81	1
GBM	11.94	8.73	3
GARCH	21.66	22.52	0
Merton	14.24	11.17	0
GARCH-Merton	15.98	12.57	0

Table S4.2 · AAPL

Model	Mean RMSE%	Median	# regimes best
MD-GBM	32.59	28.36	0
GBM	31.64	28.41	1
GARCH	39.76	30.05	1
Merton	30.60	27.26	2
GARCH-Merton	41.53	27.68	0

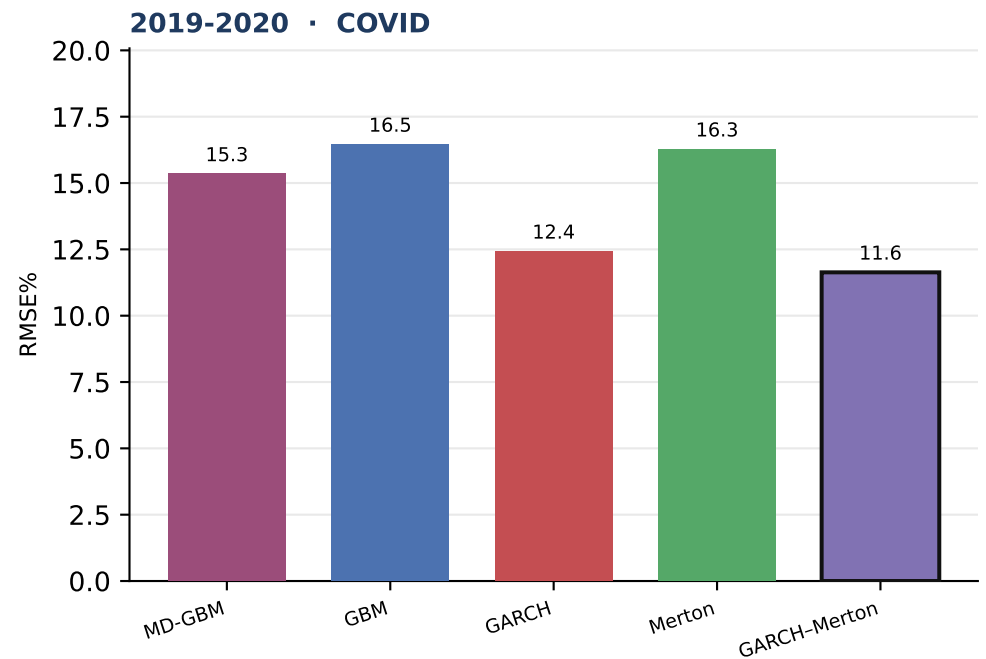
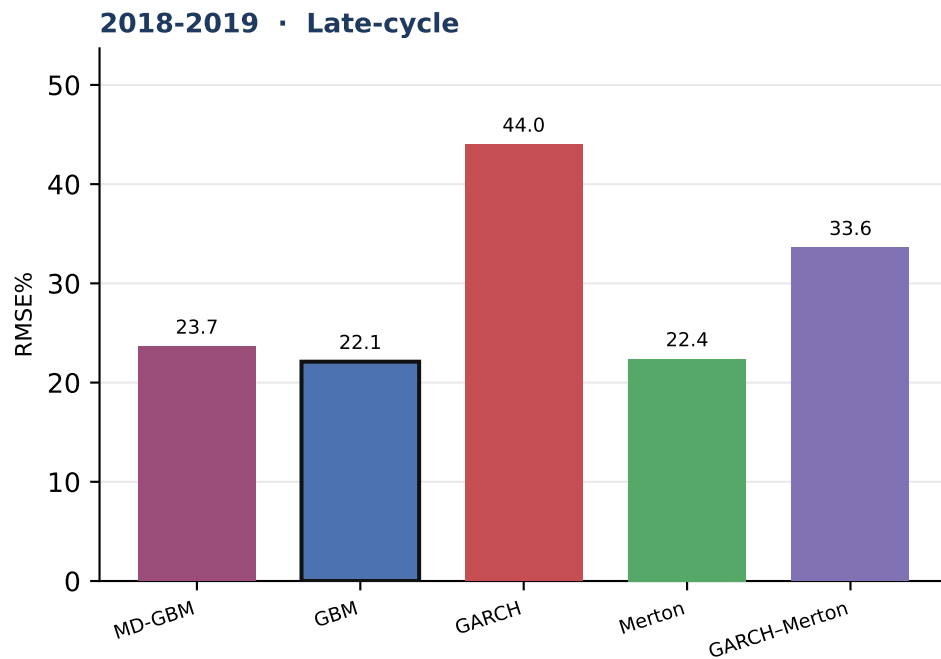
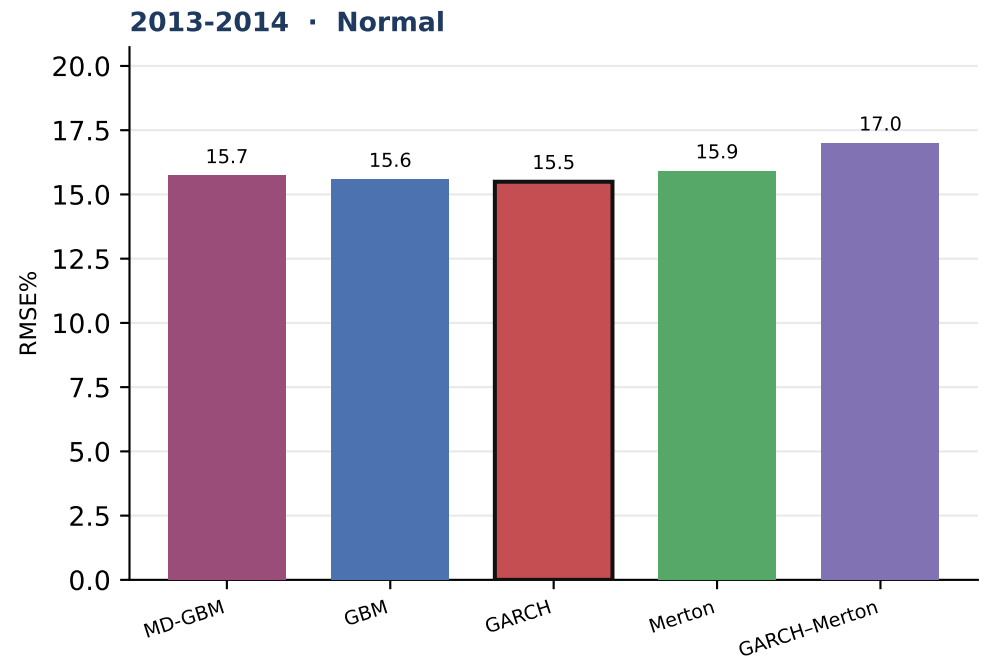
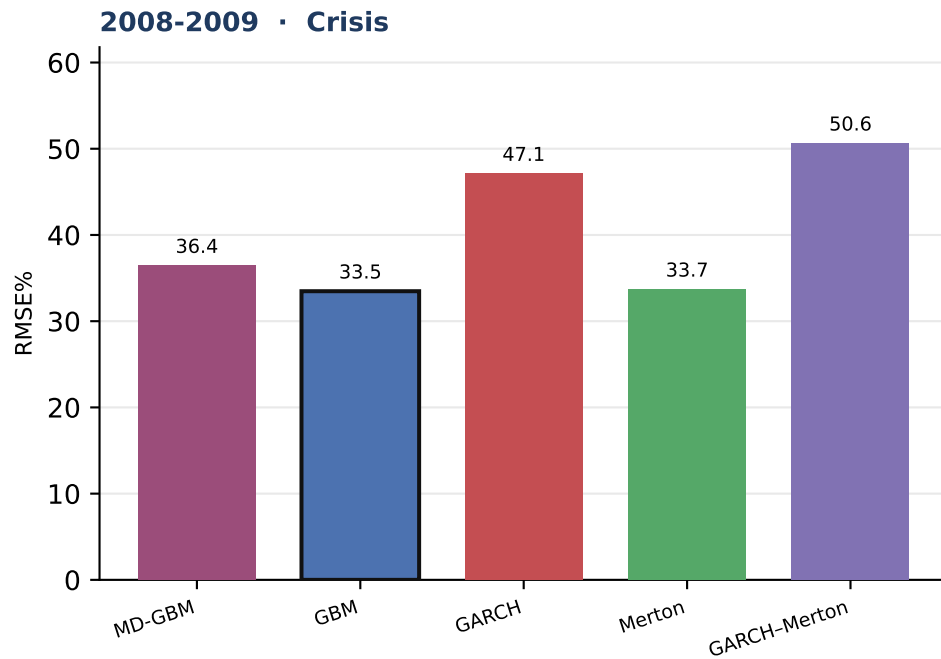
Table S4.3 · MSFT

Model	Mean RMSE%	Median	# regimes best
MD-GBM	11.41	11.39	0
GBM	11.48	11.58	0
GARCH	13.29	12.14	2
Merton	11.14	11.58	2
GARCH-Merton	13.44	12.83	0

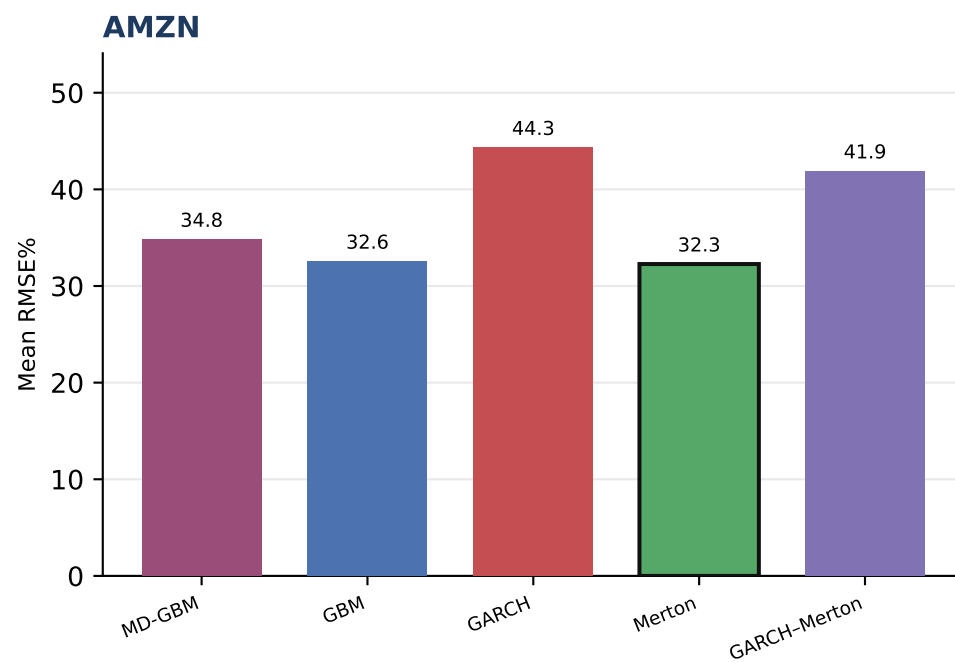
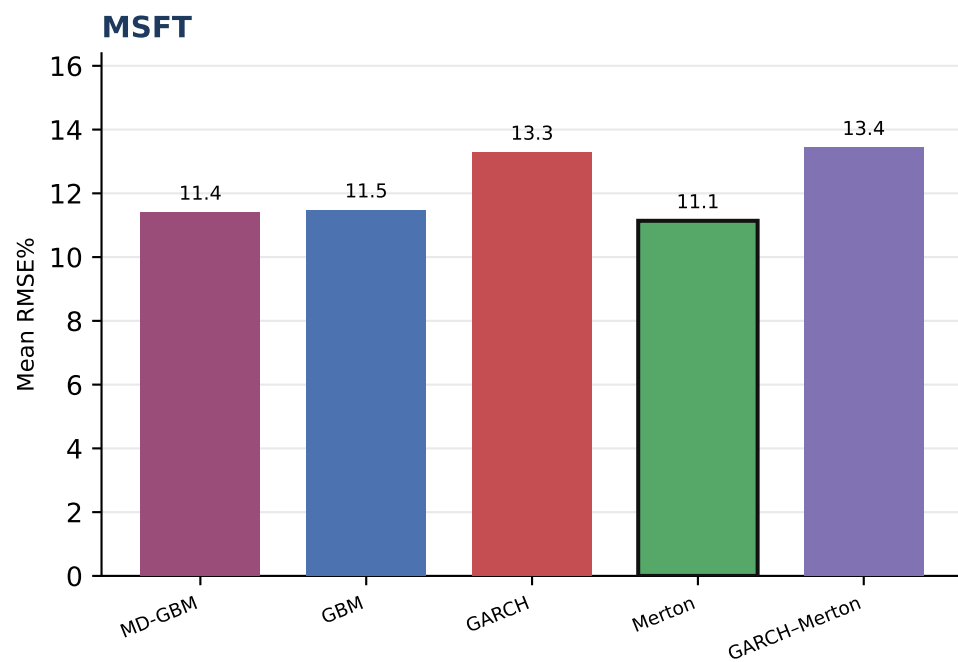
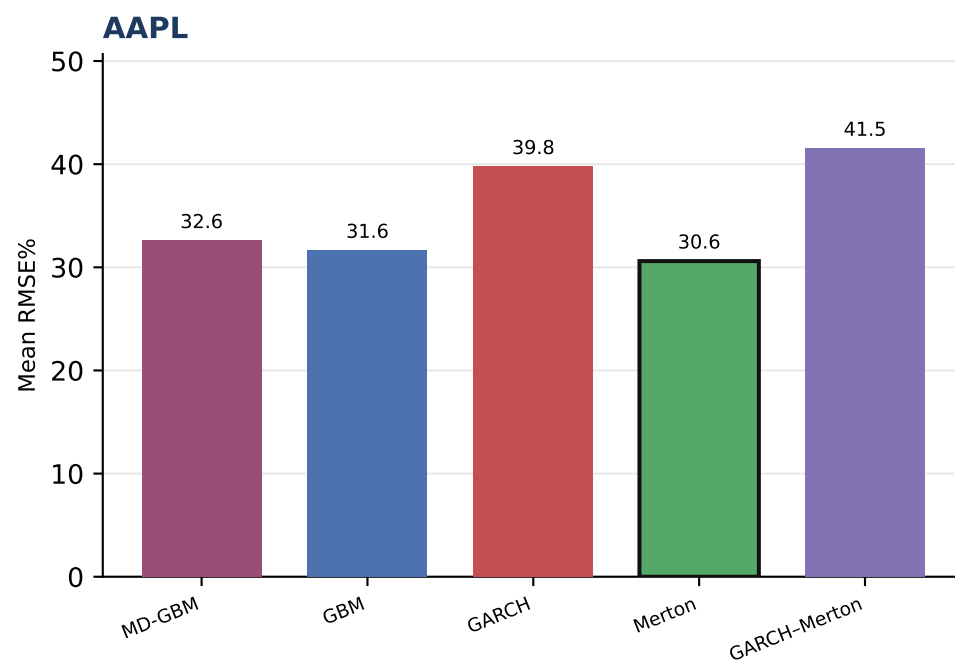
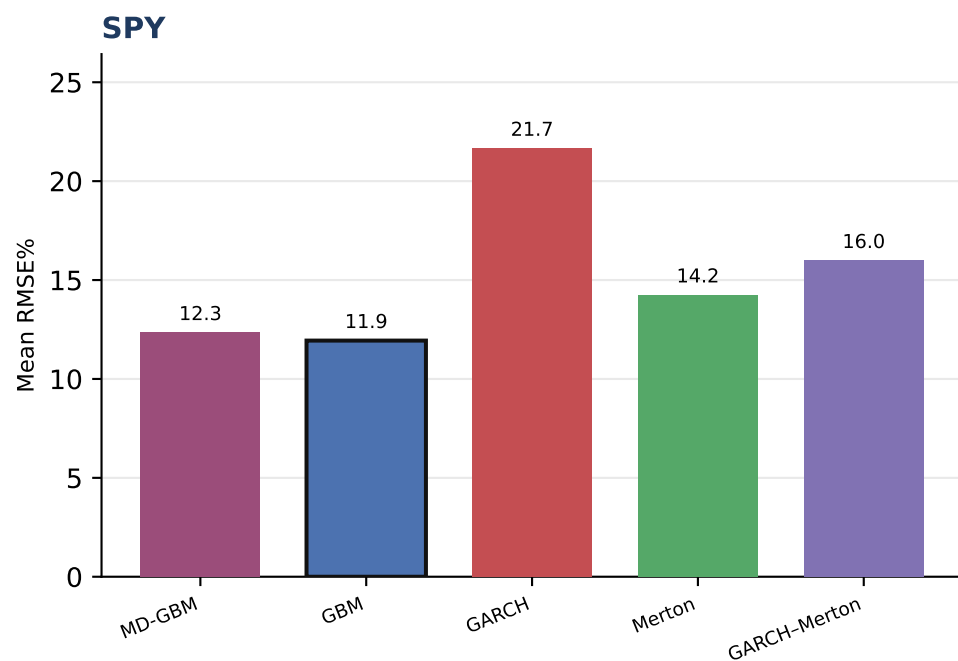
Table S4.4 · AMZN

Model	Mean RMSE%	Median	# regimes best
MD-GBM	34.83	39.10	0
GBM	32.56	35.83	1
GARCH	44.33	43.11	1
Merton	32.27	32.88	2
GARCH-Merton	41.90	44.64	0

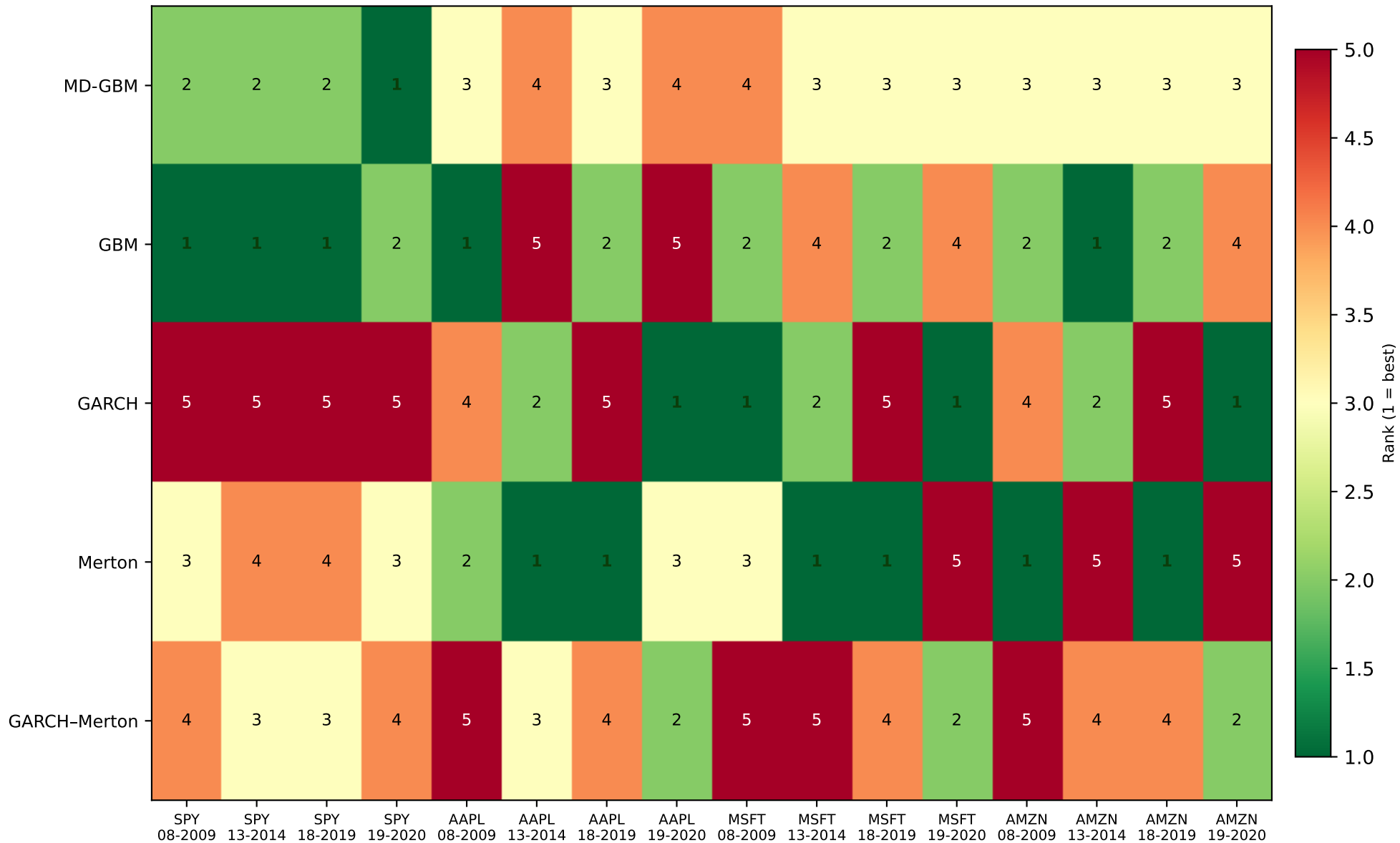
Each panel is one company. Green row = lowest mean RMSE% across the four V3 regimes for that name.



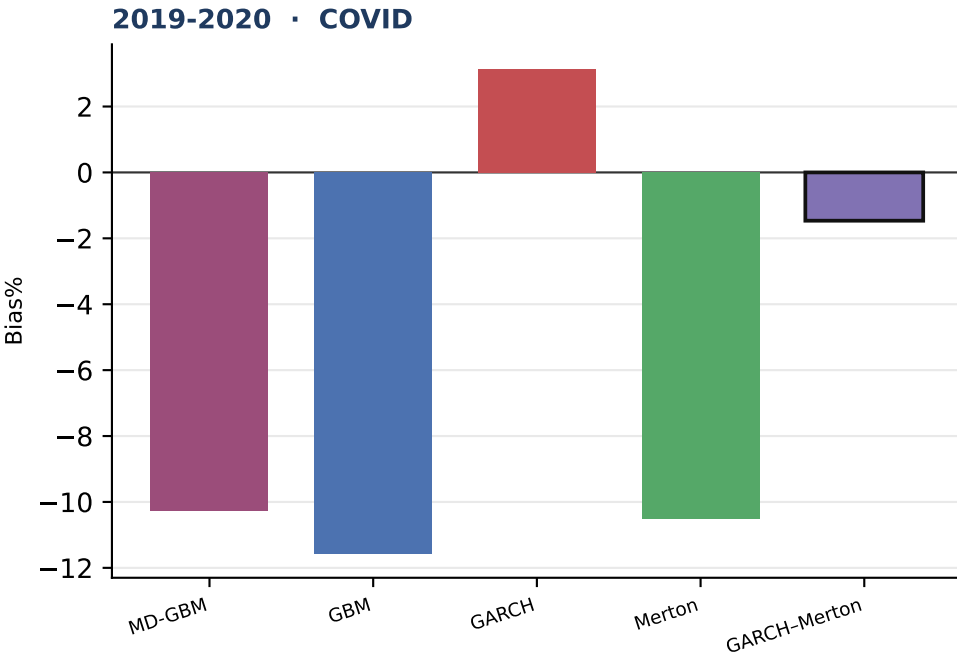
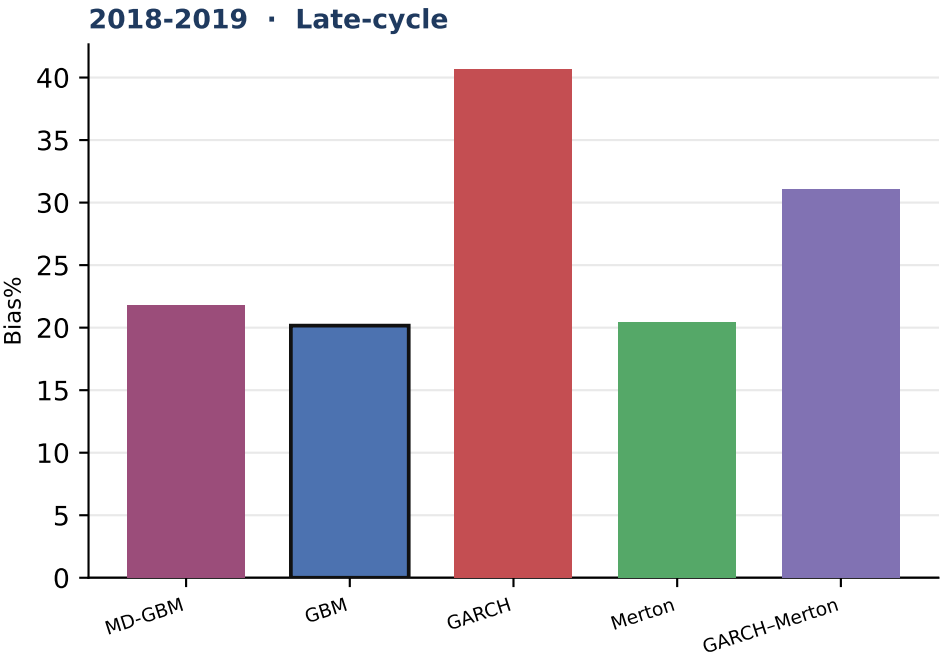
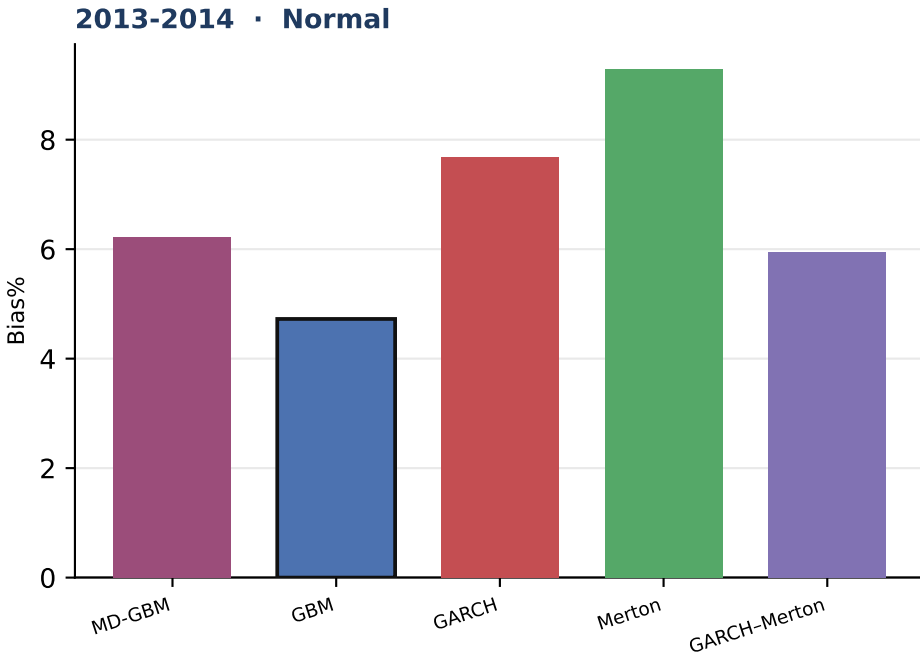
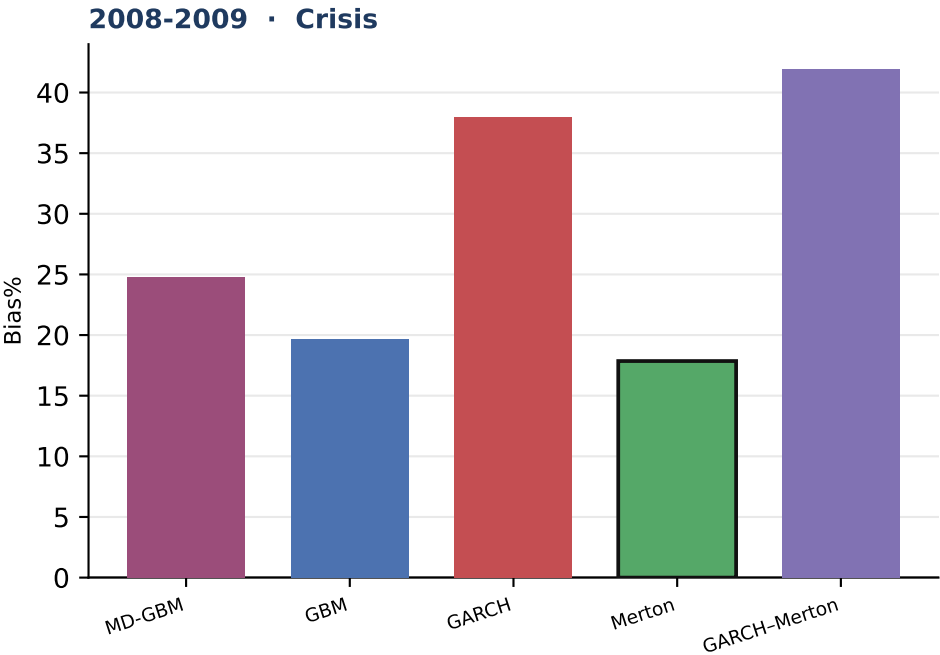
Outlined bar = lowest mean RMSE% in that regime. Equal-weight mean of the four companies.



Outlined bar = lowest mean RMSE% for that company.



Green / 1 = lowest RMSE% in that company×regime cell. Rankings can and do change across names and volatility states.



Outlined bar = closest to zero Bias% (not the RMSE ranking). Positive = model expensive vs listed quote.

Final analysis

Which models track realized S_t most closely?

- On mean RMSE% of the p50 path versus realized S_t across 16 company×regime cells, GBM is best (mean RMSE% = 21.90; 5 of 16 cells).
- The ranking is not stable. Best-in-cell models: GARCH, GBM, MD-GBM, Merton.
- These conclusions are conditional on 1.5-year monthly calibration P-measure calibration, 10000 paths, daily steps, and the p50 scoring convention.